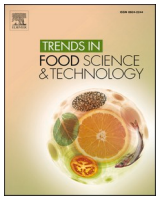
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From plate to palate: Sustainable solutions for upcycling food waste in restaurants and catering

Nida Kanwal [a,b], Min Zhang [a,c,*], Mustafa Zeb [d], Uzma Batool [a], Imad khan [a], Luming Rui [e]

[a] State Key Laboratory of Food Science and Resources, Jiangnan University, 214122 Wuxi, Jiangsu, China
[b] Jiangsu Province International Joint Laboratory on Fresh Food Smart Processing and Quality Monitoring, Jiangnan University, 214122 Wuxi, Jiangsu, China
[c] China General Chamber of Commerce Key Laboratory on Fresh Food Processing & Preservation, Jiangnan University, 214122 Wuxi, Jiangsu, China
[d] International Joint Research Laboratory for Biointerface and Biodetection, Jiangnan University, Wuxi, China
[e] Yechun Food Production and Distribution Co.,Ltd.,225000, Yangzhou, Jiangsu, China

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ABSTRACT

Background: Food waste (FW) in restaurants and catering arises from inefficiencies throughout food preparation, handling, and disposal stages. Addressing these inefficiencies via FW upcycling into value-added products presents a sustainable strategy to mitigate waste and advance environmental sustainability.
Scope and approach: This study explores the global impact of food waste, factors influencing its generation, and potential environmental benefits from waste prevention. It examines consumer food waste rates, where approximately 17.5% of prepared food is discarded, comprising kitchen waste (2.2%), serving waste (11.3%), and customer leftovers (3.9%). It explores methodologies used to estimate food waste across EU countries, comparing waste management practices in the UK and the Netherlands. It investigates food waste strategies in China, particularly in ethnic restaurants, and considers consumer perceptions regarding packaging's role in waste reduction. Additionally, the study covers sustainable waste management practices in Asia, including pretreatment techniques such as upcycling, microbial conversion, anaerobic co-digestion, and 3D food printing, aimed at reducing waste by 20–30%. Furthermore, it evaluates machine learning models for predicting catering demand with 85% accuracy and analyzes a 25% reduction trend in waste within Swedish public catering.
Key findings and conclusions: The study highlights challenges and future prospects in repurposing food waste within restaurant and catering sectors. Presently, only 10–15% of food waste is repurposed, with potential to increase to 30–40% through enhanced practices and technologies. Continued research and innovation in sustainable waste management can potentially reduce overall waste by 25%, conserve resources, and foster a more sustainable food system.

1. Introduction

Food waste refers to edible food that is discarded or left to spoil due to over-preparation, improper storage, and inefficient practices (Yahia & Mourad, 2020). Restaurants and catering businesses generate a significant amount of food waste, including surplus ingredients, trimmings, and unsold meals. This waste arises from various stages, including food preparation, service, and leftovers from customers, leading to significant economic and environmental impacts (Ishangulyyev, Kim, & Lee, 2019). Furthermore,58% of fugitive methane emissions at landfills come from wasted food, according to an updated report from the U.S. EPA (States, 20 MAY 2024). According to the Food and Agriculture Organization (FAO), approximately one-third of all food produced for human consumption is lost or wasted and 1.3 billion tons of food are wasted globally each year, with a significant portion occurring in the catering and hospitality sectors. In the United States, the Food Waste Reduction Alliance (FWRA) reports that 4–10% of food purchased by restaurants is wasted before reaching the consumer (Dahlan, Yusoff, Akinbile, Wang, & Wang, 2022). At the European level, the European Parliament's Committee on Agriculture reports that up to 50% of edible and safe food is needlessly discarded in households, supermarkets, restaurants, and throughout the entire food chain annually (EU Commission, 2024). Meanwhile, 79 million people live below the poverty line, and 16 million rely on food aid from charities. The global economic value of food waste is estimated at around $1000 billion annually, and this figure increases to $2600 billion when accounting for the hidden

* Corresponding author. School of Food Science and Technology, Jiangnan University, 214122 Wuxi, Jiangsu Province, China.
E-mail address: min@jiangnan.edu.cn (M. Zhang).

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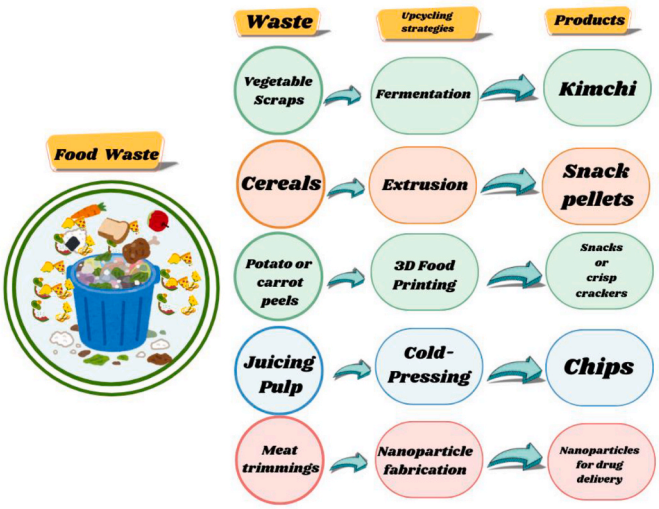


Fig. 1. Innovative food waste upcycling approaches and products.

environmental costs associated with the issue (Waste, 2019). Approximately 44%–47% of annual fruit, vegetable, meat, and fish production is wasted (Aureli, Scalvedi, & Rossi, 2021). Waste and Resources Action Programme (WRAP) in the UK found that the hospitality and food service sector produces around 1 million tons of FW annually, costing the industry an estimated £2.5 billion. In Europe, Food Use for Social Innovation by Optimizing Waste Prevention Strategies (FUSIONS) project estimated that food waste in the hospitality sector amounts to approximately 12.5 million tons per year (WRAP, 2024). Additionally, a study by Leanpath indicates that on average, 4–8% of food in the catering sector is wasted, resulting in significant financial losses (Leanpath, 2024). The National Restaurant Association in the US highlights that food waste can account for up to 10% of a restaurant's food cost, impacting overall profitability. Contributing factors include inappropriate procurement, storage, processing, and operation strategies (Elgarahy et al., 2023b). In the restaurant and catering industry, both staff and consumers are major contributing factors to food waste through over-preparation, improper handling, and uneaten leftovers. This waste not only contributes to environmental degradation but also represents a missed opportunity to address food insecurity and reduce greenhouse gas emissions (Rakesh & Mahendran, 2023).

The need for sustainable transformations has never been more urgent and the need to review the topic of food waste in the restaurant and catering sectors is crucial, especially as the world population is projected to reach around 8.5 billion by 2030 according to a 2015 UN report (Elgarahy et al., 2023b). Although food-related waste in this sector is frequently covered in the media, it has not yet received adequate academic attention. With the global population already facing food poverty, addressing the 12–14% contribution of food waste by these sectors is imperative. Reducing food waste can play a significant role in enhancing food security, conserving resources, and mitigating environmental impacts, making this an urgent area for research and intervention. One innovative and sustainable approach is the upcycling of food waste, which can contribute to stable ecosystems and a circular economy. Implementation of sustainable menu planning is the key strategy for upcycling food waste in restaurants and catering. By designing menus that utilize ingredients efficiently and creatively, chefs can minimize waste while offering innovative and appealing dishes to customers (Rohmer, Gerdessen, & Claassen, 2019). Upcycling FW in restaurants and catering is not only a practical necessity but also a moral imperative (Aleshaiwi, 2023). Through collaboration, education, and innovation, the transition from plate to palate can become a model for sustainable food practices worldwide (Brennan et al., 2021; Rohmer et al., 2019).

The objective of this article is to examine the main causes of FW in



Fig. 2. Catering food waste causes.

the restaurant and catering sectors across different regions of the world, its environmental, social, and economic impacts, and innovative approaches to upcycling FW into useful by-products, emphasizing sustainable practices for waste reduction and promoting a circular economy. Furthermore, integrated analysis of factors influencing food waste, including consumer behavior, management perspectives, and technological advancements. While previous studies have addressed various aspects of food waste, there is a lack of exploration of the main contributing factors in these sectors. Additionally, the scope of food waste is often not compiled in a manner that supports the development of upcycling strategies. Therefore, this study conducts a comprehensive analysis of the causes and effects of food waste, quantifying the data of loss worldwide through a thorough literature review, and exploring innovative upcycling strategies. These strategies include high-pressure processing, fermentation, microbial application, innovative packaging, 3D food printing, and machine learning, along with chemical, biological, and physical methods, accompanied by practical examples. However, information on food waste and its upcycling strategies in the restaurant and catering sectors remains scattered and limited, and public acceptance of upcycled products is not widespread, posing a significant challenge for the adoption of these strategies. Therefore, the authors attempt to explore various strategies for upcycling FW (Fig. 1) into economically valuable products such as food items, conversion into green energy, providing an overview of recent innovations, benefits, and technical challenges involved. It analyzes common resources found in highly cited literature reviews from recent years, encompassing a total

of 140 published articles from 2017 to 2024, offering a robust basis for advancing sustainable food waste management practices.

2. Contributing factors and global impacts of food waste

Globally, it is estimated that about 12–14% of FW occurs in the food service sector, which includes restaurants and catering services (Xue et al., 2017). Factors (Fig. 2) contribute to this FW which need to be clearly articulated to understand their impact have been explored in this paper. One significant challenge is predicting customer demand, leading to over-preparation and wastage. The absence of a structured purchasing plan exacerbates this issue, resulting in food items being forgotten and subsequently spoiling (Filimonau & Gherbin, 2017). Excessive stocking of ingredients further adds to the problem, often causing surplus food to go unused. Additionally, compliance with strict regulations often results in discarded food, while residual waste from food preparation is a common occurrence due to trimming and peeling (Filimonau, Fidan, Alexieva, Dragoev, & Marinova, 2019). Inadequate equipment and tools can also hinder efficient food usage, and uncertainty about the required cooking quantities often leads to over-production (Filimonau et al., 2019). Portion sizes are another critical factor; serving inappropriate portions can result in significant leftovers. Buffet-style service and extensive menus contribute to food waste by promoting variety over necessity, leading to excess food preparation (Malefors, 2021). This often results in significant amounts of uneaten food being discarded, as it cannot be reused or safely stored for future use. Inefficient waste management practices and poor facility layouts further aggravate the issue by not effectively handling waste. Staff handling food improperly and communication breakdowns among staff are also significant contributors to food waste (Filimonau & Delysia, 2019). Impulse purchasing, varied dietary preferences, and the perception of food as unappealing can lead customers to order more food than they consume, resulting in wastage. The United Nations Environment Program (UNEP) reports that factors such as purchasing power, industrial expansion, and diverse food offerings contribute to the substantial volume of food waste generated in Asian regions (Withanage, Dias, & Habib, 2021). Building on the contributing factors, it is essential to consider the impacts of food waste, which play a significant role in aggravating these issues. The environmental consequences of FW are significant, contributing up to 8% of total global greenhouse gas emissions. Methane, a potent greenhouse gas, aggravating climate change (Dhir, Talwar, Kaur, & Malibari, 2020). FW also increases landfill waste, releasing harmful substances during decomposition (Schanes, Dobernig, & Gözet, 2018). Moreover, food production consumes substantial water and land resources, so wasting food also wastes these resources used for production, transport, and preparation, such as water, energy, and labor. In addition to the environmental consequences, the economic implications of FW in the restaurant and catering sector are equally profound, affecting both operational efficiency and profitability, representing a loss of valuable resources and money for businesses (Filimonau, Todorova, Mzembe, Sauer, & Yankholmes, 2020). This inefficiency directly affects profitability and sustainability, as businesses incur costs without receiving the full benefit of their investment. FW also leads to higher disposal costs. By reducing FW, businesses can improve their bottom line, increase sustainability, and use resources more efficiently (Slorach, Jeswani, Cuéllar-Franca, & Azapagic, 2019). Moreover, the social and health impacts of FW in the restaurant and catering sector cannot be overlooked, as they directly affect community well-being and public health, especially considering global food insecurity and hunger (Malefors, 2022). Wasted edible food represents a missed opportunity to alleviate hunger and malnutrition, aggravating food scarcity issues. Additionally, food waste can have social implications regarding equity and access to food. Proper handling of food waste is crucial, as improper disposal can lead to the spread of foodborne illnesses and contamination of water sources, affecting both human health and the environment. Managing food waste effectively can reduce health risks and contribute to a healthier



Fig. 3. Upstream and downstream intervention for food waste management.

environment. Furthermore, FW in the restaurant and catering sectors can reflect cultural attitudes towards food consumption and wastefulness. Addressing these cultural norms and promoting sustainable practices can help reduce food waste and shift societal attitudes towards food (Phasha et al., 2020). However, addressing the issue of food waste requires a multifaceted approach that includes both upstream and downstream interventions (Fig. 3) (Malefors, Sundin, Tromp, & Eriksson, 2022a). Also it requires investments in infrastructure development and improvements in supply chain management practices to ensure efficient storage, transportation, and delivery of food, reducing the risk of spoilage and waste (Elgarahy et al., 2023b). Lastly, regulatory factors including food safety regulations and incentives for waste reduction, play a role in food waste generation and require a balance between food safety and waste reduction (Economou et al., 2024).

3. Ecological and economic benefits of upcycling food waste

Upcycling FW in the restaurant and catering sector offers significant potential environmental advantages. By transforming food scraps and surplus into new, value-added products, this practice reduces the amount of waste sent to landfills, thereby lowering greenhouse gas emissions such as methane, a potent contributor to climate change (Elgarahy et al., 2023b). Additionally, food waste can be converted into green energy, providing an alternative to fossil fuels (Soon, 2023). This is particularly beneficial for countries like Pakistan, which faces severe electricity shortages. Moreover, FW is rich in nutrients that can be upcycled into appealing and attractive food products through innovative technologies such as 3D food printing (Iris & Wong, 2023; Guo, Zhang, & Bhandari, 2019). This sustainable approach not only enhances resource efficiency but also fosters a circular economy, promoting a more sustainable food system and reducing the environmental footprint of the food service sector. To implement this, it is important to know some facts, for example each year the European Union throws away about 88 million tons of food, which makes up 15–16% of the total environmental impact of its food value chain (European Commission, 2023).

However, implementing food waste reduction initiatives in these sectors can yield significant financial benefits. For example, research conducted by the World Resources Institute (WRI) and the Waste and

Table 1
Upcycled food waste objectives and relevant research.

	Objectives	Research findings	References
1	Non-pomace grape juice residues	The polyelectrolyte complexation seems effective due to the upcycling of the grape juice non-pomace residues in the complexes which influenced the functionalization of the physiochemical properties of the freshness indicators	de Azevedo and Noreña (2023)
2	Pig skin waste (gelatin)	The three-dimensional network results in gel strength, making the gelatin a commercial product for the food industries	(LATORRE, Velazquez, Palacio, & Sanchez, 2022)
3	Upcycled dietary fibre	Upcycled DF from persimmon byproducts as a potential modulator agent of microbial populations and showed the antibacterial effects of gallic acid after release	Moreno-Chamba et al. (2022)
4	Consumer purchase intention towards upcycled food	The highest purchase intention belongs to younger female consumers who frequently recycled at home and had a higher quality and better taste expectation from upcycled food	Yilmaz and Kahveci (2022)
5	Fish by-products	60–100% of the total oil content recovered within 19 min from the fish by-products, whereas 51% MUFAs, 29% PUFAs and 20% SFA were included and a profile of fewer cardiovascular risks, less microbial growth	Pinela et al. (2022)
6	Muffins	15% muffins were the most similar to the control with improved fibre content, mineral content, amino acid profile, and antioxidant activity	Grasso, Pintado, Pérez-Jiménez, Ruiz-Capillas, and Herrero (2021)
7	Sustainable development in the food sector	The mandatory six transformations based on sensory-derived food satisfaction, wellbeing and happiness include diet shift, food diversity, reduced food waste, circular food system, prioritizing well-being through the foods and tolerating the climatic outcomes	Aschemann-Witzel, Ares, Thøgersen, and Monteleone (2019)
8	Circular Bio-economy Opportunities	Circular bioeconomy enhances the value of material flow in an eco-friendly way where the driving force is the increasing energy requirement	Mak, Xiong, Tsang, Iris, and Poon (2020)
9	Upcycled orange fibre (Gluten free muffins)	Microwave technology produced muffins with a higher height, easier to chew, and better appearance than conventional baking, independently of the fibre used	Rodríguez, Alvarez-Sabatel, Ríos, Rioja, and Talens (2022)
10	As balanced Diet	Upcycling food contributes to increasing resource use efficiency (SDG 12.5) and substantially reduce waste generation, ensuring sustainable food consumption and production systems (SDG 12.5; SDG 12.3)	Nogueira, Alves, and Vaz-Fernandes (2021)
11	Circular economy by circular food behavior	The evolution of linear behaviors, going through slightly more transitioning behaviors until reaching circular behaviors	(do Canto, Grunert, & De Barcellos, 2021)
12	Food management hierarchy (position of upcycled food)	In the food waste management hierarchy, upcycled food production can be included in the hierarchy, it appear below the redistribution and above the animal feed stage of the hierarchy	Moshtaghian, Bolton, and Rousta (2021)

Resources Action Programme (WRAP) evaluated cost and benefit data from 1200 business sites across 700 companies in 17 countries. They found that for most businesses, every $1 invested in food waste reduction resulted in savings of $14 or more (EU Commission, 2024). Therefore, it can be inferred that SDG 12.3, set by the United Nations, aims to cut global per capita food waste at retail and consumer levels in half by 2030 (Ardra & Barua, 2022; Jacob-John, D'Souza, Marjoribanks, & Singaraju, 2023) is attainable and could be exceeded with initial investments, managements, and its beneficial for ecology and economy. Another study also investigated the feasibility of SDG 12.3 and assessed the environmental benefits, including those for climate, biodiversity, and overall environment, of preventing food waste in the food service sector (Beretta & Hellweg, 2019). It highlights the importance of individual food services in achieving this goal, noting that customized measures are necessary based on factors such as serving systems, customer segments, and preparation methods. Further case studies are needed to establish parameters for different types of food services and determine the most effective food waste reduction strategies. From practical perspective Table 1 presents different upcycled food waste objectives and relevant research.

3.1. Five key approaches for eco-economic benefits from food waste

Restaurants and catering businesses can play a crucial role in FW management by adopting sustainable practices within a comprehensive value framework. Five key approaches; repositioning, reallocating, reacting, re-engineering, and relating, highlight how these businesses can achieve economic, social, and environmental benefits (Huang et al., 2021).

Repositioning involves increasing the visibility and likelihood of consumption of dishes or ingredients nearing expiration (Sundström, 2021). For example, offering discounts or creating special menu items from soon-to-expire ingredients can reduce waste and recover costs. Reallocating focuses on redistributing surplus food to those in need, often through partnerships with local food banks or charities, addressing food insecurity and enhancing social responsibility (Sun et al., 2024). Reacting involves immediate measures such as reducing prices on items close to expiration or donating unsold food to charities, but long-term solutions are essential. Re-engineering includes redesigning processes or products to minimize waste, such as adjusting portion sizes,

optimizing inventory management, or using innovative packaging solutions to extend shelf life. These changes can significantly reduce environmental impact and improve profitability (Krishnan, Agarwal, Bajada, & Arshinder, 2020). Furthermore, relating involves building strong relationships with customers, suppliers, and stakeholders to promote sustainable practices. Educating customers about reducing food waste, collaborating with suppliers for sustainable sourcing, and engaging staff in waste reduction are key. These efforts can contribute to a more sustainable food system. However, the impact of these benefits varies, indicating a need for further empirical investigation. Adopting a holistic approach that considers the entire supply chain can help these businesses reduce waste, improve their bottom line, and support a more sustainable future (Riesenegger & Hübner, 2022).

4. Sustainable solutions for upcycling food waste in restaurants and catering

4.1. Turning scraps into culinary art: the future of food waste with 3D printing

3D food printing is an emerging technology that is gaining significant attention, with increasing research and development focused on its potential to revolutionize food production (Zhao, Zhang, Chitrakar, & Adhikari, 2021). This technology offers innovative solutions for food waste by transforming discarded material into valuable products. For instance, in a study it was analyzed that the administration of grape pomace can significantly enhance the nutritional and antioxidant properties of cookies. Using specific parameters such as a nozzle diameter of 1.28 mm, extruder motor speed of 600 rpm, and print speed of 400 mm/min optimized the printability of the food using the CARK extrusion-based 3D printer. Post-processing at 130 °C for 12 min further refined the printed constructs, with the 6% grape pomace formulation proving most favorable during sensory evaluation (Jagadiswaran et al., 2021). Notably, the final product was enriched with proteins and dietary fiber. This approach underscores the potential for adding value to industrial waste streams while meeting consumer preferences. It also highlights how additive manufacturing allows for precise customization of food products in terms of nutritional content, promoting cleaner production practices and enhancing resource recovery from food processing wastes (Jagadiswaran et al., 2021).

By repurposing food waste into familiar consumer products like cookies, chocolates (Karyappa & Hashimoto, 2019), biscuits, and wafers, 3DFP can contribute to sustainability efforts. For instance, vegetables with high starch content are generally more amenable to 3D printing. A study by Pant, Ni Leam, Chua, and Tan (2023) characterized food ink recipes made from spinach and kale stalks, discovering that the formulation of the ink depended significantly on the inherent water content of the processed material. Specifically, they found that developing inks from spinach was more challenging than from kale due to its higher water content. Furthermore, the study highlighted the potential of 3D printing to transform food waste into visually appealing edible products. For example, Leo et al. (2022) utilized 3D printing to convert orange peel waste (OPW) into edible snacks rich in bioflavonoids. Their research produced various shapes using OPW inks, including edible soup bowls, biscuits, and food toppings, all of which served as sources of antioxidant-rich compounds such as gallic acid, p-coumaric acid, ferulic acid and narirutin. This demonstrates the viability of using 3D printing technology to upcycle food waste into nutritious and aesthetically pleasing products in the restaurant and catering sectors (Leo et al., 2022; Pant et al., 2023).

4.2. Machine learning models and food waste reduction

According to a study machine learning models were designed to enhance food demand forecasting and to improve accuracy, thereby aiding in the avoidance of overproduction or underproduction. Additionally, two baseline models were presented to compare against traditional demand estimation methods. The effectiveness of the proposed machine learning models was evaluated using data from three canteens. Results indicated that models based on random forest (RF) and long short-term memory (LSTM) neural network algorithms provided the most precise forecasts. Implementing these advanced models could potentially reduce wasted meals by 14%–52% and decrease unmet demand by 3%–16% compared to the baseline models. The integration of machine learning into demand forecasting has the potential to enhance service levels and reduce food waste, thus addressing its environmental, social, and economic impacts (Rodrigues, Miguéis, Freitas, & Machado, 2024).

Moreover, the RF algorithm-based model performed better in the school canteens, while the LSTM neural network-based model showed superior performance in the company canteen. The RF model showed higher effectiveness with three to four years of data and 200 to 400 daily meals, whereas the LSTM model excelled with over nine years of data and approximately 1900 meals served daily. LSTM models exhibited the most significant potential for FW reduction, particularly in FCS3, likely due to a longer data history for more accurate predictions. Machine learning-based forecasts consistently outperformed traditional methods, with RF and LSTM models yielding the best results. FCSs not utilizing machine learning should consider adopting a Moving Average (MA) approach over a naive method for demand forecasting (Yuan et al., 2023).

5. Global perspectives on food waste upcycling in restaurant and catering industries

The percentage of FW in restaurants and catering sectors varies significantly depending on the region, type of establishment, and management practices. Globally, it is estimated that about 12–14% of food waste occurs in the food service sector, which includes restaurants and catering services (UNEP, 2021). In the United States, the restaurant sector is responsible for approximately 15–20% of total FW generated (Gunders & Bloom, 2017). In the European Union, the hospitality sector, encompassing restaurants, hotels, and catering services, accounts for around 12% of total FW (Papargyropoulou et al., 2019). Specifically, in the United Kingdom, it is estimated that about 18% of the food purchased by the hospitality sector is wasted, equating to roughly 920,000 tonnes of FW annually (UK, 2024). In 2019, global FW varied significantly across different sectors. The food service sector contributed an average of 32 kg per capita per year, totaling 244 million tons. The retail sector accounted for 15 kg per capita per year, amounting to 118 million tons. Households were responsible for the highest average, with 74 kg per capita per year, equating to 569 million tons. Overall, the total global food waste was 121 kg per capita per year, reaching 931 million tons. These figures highlight the significant contribution of the restaurant and catering sectors (D. UNEP, 2021).

5.1. Restaurants in UK and Netherlands

A study by Filimonau et al., adopted a qualitative and descriptive case study approach. This was deemed the most suitable method due to the exploratory and sensitive nature of the topic, specifically restaurant food waste. The research focused on the experiences of restaurant managers in the UK and the Netherlands, shedding light on food waste management practices in these two countries. The study's scope was limited to these regions due to resource constraints for cross-national comparison (Filimonau et al., 2020). This comparative study revealed shared practices like demand forecasting, yet inefficiencies persist, leading to substantial waste. Financial concerns often hinder proactive measures like surplus food redistribution, despite environmental benefits. Effective strategies in British and Dutch restaurants include demand forecasting and passive disposal, but proactive measures such as repurposing ingredients and reducing plate waste are less common.

Table 2
SWOT analysis of food waste treatment methods.

Strengths	1 Environmental benefits: FW treatment methods contribute to reducing greenhouse gas emissions, minimizing landfill usage, and conserving natural resources (Al-Rumaihi, McKay, Mackey, & Al-Ansari, 2020). 2 Resource recovery: Many FW treatment methods facilitate the extraction of valuable resources such as biogas, compost, and biofuels, which can be used for energy production or agricultural purposes (P. Sharma et al., 2020). 3 Regulatory support: Governments and environmental agencies increasingly support and incentivize FW treatment initiatives through regulations, subsidies, and grants (Joshi & Visvanathan, 2019). 4 Technological advancements: Continuous innovation in FW treatment technologies improves efficiency, scalability, and cost-effectiveness, making them more viable and accessible (Ashokkumar et al., 2022). 5 Public awareness: Growing awareness of environmental issues and sustainable practices enhances public acceptance and participation in FW treatment programs (Rasool, Cerchione, Salo, Ferraris, & Abbate, 2021).
Weaknesses	1 Infrastructure requirements: Implementing FW treatment methods often demands significant investments in infrastructure, such as processing facilities, collection systems, and transportation networks (Babbitt et al., 2022). 2 Operational challenges: Handling and treating diverse types of FW can present operational complexities, including odor control, contamination risk, and seasonal variations in waste generation (Pour & Makkawi, 2021). 3 Economic viability: The economic feasibility of FW treatment methods depends on factors like waste composition, market demand for by-products, and regulatory frameworks, posing challenges for profitability and scalability (Dou, Dierenfeld, Wang, Chen, & Shurson, 2024). 4 Technological limitations: Some food waste treatment technologies may have limitations in processing certain types of waste or achieving optimal resource recovery rates, hindering their effectiveness. 5 Behavior change: Addressing FW at its source requires significant shifts in consumer behavior, food industry practices, and supply chain management, presenting ongoing challenges in adoption and sustainability (Aschemann-Witzel & Stangherlin, 2021).
Opportunities	1 Circular economy initiatives: Food waste treatment aligns with the principles of the circular economy, offering opportunities for creating value FW streams, closing resource loops, and promoting sustainable consumption and production (Rashid & Shahzad, 2021). 2 Innovation and collaboration: Collaborative efforts among government, industry, academia, and NGOs can foster innovation in FW treatment technologies, business models, and policy frameworks, unlocking new opportunities for efficiency and effectiveness (Annosi, Brunetta, Bimbo, & Kostoula, 2021). 3 Market development: Growing demand for sustainable products and services creates market opportunities for businesses involved in FW treatment, including technology providers, waste management companies, and renewable energy producers. 4 Community engagement: Engaging communities in FW reduction and treatment initiatives through education, incentives, and participatory programs can enhance local resilience, social cohesion, and environmental stewardship (Aschemann-Witzel & Stangherlin, 2021). 5 Policy support: Supportive policies, such as landfill bans, extended producer responsibility schemes, and tax incentives, can create favorable conditions for investment in FW treatment infrastructure and innovation (Fesenfeld, Rudolph, & Bernauer, 2022).
Threats	1 Economic uncertainties: Economic downturns, market volatility, and financial constraints may limit investments in food waste treatment infrastructure and innovation, slowing progress in addressing food waste (Peschel & Aschemann-Witzel, 2020). 2 Regulatory challenges: Inconsistent regulations, bureaucratic hurdles, and policy uncertainties can create barriers to implementing and scaling up FW treatment initiatives, complicating planning and decision-making. 3 Competing priorities: Other pressing environmental, social, and economic issues may divert attention and resources away from FW reduction and treatment efforts, diluting momentum and support (Al-Rumaihi et al., 2020). 4 Technological risks: Dependence on specific technologies or reliance on unproven innovations may expose treatment FW initiatives to technical failures, performance setbacks, and implementation delays (Cosbuc, Ungureanu-Comanita, & Gavrilescu, 2021). 5 Behavioral inertia: Overcoming entrenched habits, attitudes, and cultural norms related to food consumption, disposal, and waste management represents a formidable challenge in achieving widespread adoption of sustainable practices and behaviors (Aschemann-Witzel & Stangherlin, 2021; Bangar et al., 2024).

However, overcoming organizational and regulatory barriers is crucial to fostering sustainability. Future research needs to explore innovative methods to repurpose leftovers and optimize waste reduction strategies across the EU and globally.

5.2. Ethnic cuisine restaurants

A study by Filimonau et al., investigated ethnic food restaurants specializing in Chinese cuisine, given its popularity in the UK market of out-of-home food consumption. According to Mintel (2020), 34% of UK residents regularly eat out at Chinese restaurants, and 48% routinely order Chinese takeaways. This has made Chinese cuisine the most popular type of oriental cuisine in the UK, with 94% of customers of oriental restaurants preferring those with a Chinese specialization. Within a year of business operations, it is estimated that the case study restaurant generates 14.75 tonnes of food waste. In 2018, the restaurant catered to approximately 107,000 guests, and assuming this pattern remained unchanged in 2019, the food wastage per guest equates to 138 g (Filimonau, Nghiem, & Wang, 2021). The study also identified barriers to waste reduction and proposed strategies to address them. Future research could expand to explore food waste in restaurants specializing in other ethnic cuisines and compare waste management practices across different culinary traditions. Comparative analyses between local markets and countries of origin for ethnic cuisines could provide valuable insights. Additionally, studying FW variations among different customer segments, such as regular patrons versus tourists, and expatriates versus local residents, would deepen our understanding of waste drivers. Research efforts need to focus on promoting more efficient cooking practices among chefs and examining suppliers' roles in waste reduction across ethnic restaurants and the broader food supply chain (Filimonau et al., 2021; Wu, Zhang, Wang, Mothibe, & Chen, 2012).

5.3. Awareness in Chinese restaurants

In China, where ecological civilization and the circular economy are prioritized, effective recycling of FW faces challenges. A study utilizing an assessment model found that 37.33% of restaurant owners exhibit high awareness of food waste issues, while 62.67% show low awareness, highlighting a general lack of awareness among owners in the region. Logistic regression analysis identified factors influencing awareness, revealing that female and older owners tend to have lower awareness levels, whereas younger owners show higher awareness. Furthermore, education level, restaurant scale, daily FW generation, waste treatment methods, and city macroeconomic development were identified as factors positively influencing awareness levels (Lang et al., 2020). The rapid growth of China's restaurant industry and dining-out culture has increased the complexity of managing FW in restaurants (Lang et al., 2020).

5.4. Policy and technological drivers in Asia and the EU

In Asia, the management of FW is shaped by varying levels of awareness, development goals, and socio-economic constraints. While there is a considerable emphasis on segregating and treating FW, there is a notable gap in efforts to prevent food waste at its origin. Table 2 highlights this issue by presenting a **SWOT analysis** of FW, illustrating the strengths, weaknesses, opportunities, and threats related to current policies and practices. Furthermore, treatment methods indicates that anaerobic digestion is favored for its cost-effectiveness, energy generation, and production of nutrient-rich digestate (Giuseppe, Emanuele, Rita, Roberta, & Biagio, 2020).

Furthermore, decentralized anaerobic digestion systems are preferred over centralized ones due to their lower energy consumption, ease of operation, reduced resource requirements, lower costs, and greater public acceptance. Policies that promote energy recovery from segregated food waste play a critical role in encouraging anaerobic digestion and promoting sustainable food waste management practices (Joshi & Visvanathan, 2019; Vizzoto, Testa, & Iraldo, 2021). Studies conducted at the national level in EU countries show significant discrepancies in FW quantification due to various measurement approaches. To address this challenge, the European Commission has introduced a standardized methodology and minimum quality standards for measuring FW. However, given the diverse quantification strategies used by EU countries, there is a need for a harmonized modeling system to accurately estimate FW (Stenmarck et al., 2016). For example, let's examine a modeling methodology employed for estimating food waste across EU nations. It employs Material Flow Analysis (MFA) alongside statistical data related to food production and trade. But Implementing MFA requires extensive and detailed data collection across various stages of the supply chain. This can be time-consuming and resource-intensive, particularly in regions with fragmented or incomplete data systems. Another methodology estimates food waste using waste statistics but the challenge is waste statistics may not capture the full extent of food waste, as they often rely on reported or recorded data, which can be incomplete or inconsistent. This methodology might overlook unreported waste or informal disposal practices, leading to potential underestimations. So to overcome these challenges more research work, surveys and efforts are required to estimate exact percentage of FW to combat hunger and meet the zero hunger goal (Tchonkouang, Onyeaka, & Miri, 2023).

6. Strategies for upcycling food waste

There is no universal method for upcycling of FW (Ooi, Woon, & Hashim, 2021). Efficient FW management includes transforming it into valuable products like bio-plastics, bioenergy, enzymes, bio-char, mushroom bio-products and medical components (Guo, Zhang, & Fang, 2022; M. Sharma, Sridhar, Gupta, & Dikkala, 2022).

The United Nations Environment Program (UNEP) has estimated an economic loss of approximately 400 billion USD due to FW (Kumar et al., 2022), underscoring the necessity to utilize FW for value-added product production and to foster a circular bio economy (Rakesh & Mahendran, 2023). Transforming food waste into valuable products such as bio-plastics, bioenergy, enzymes, bio-char, and medical components is a key strategy in this regard. By converting food waste into these products, the volume of waste ending up in landfills is significantly reduced, leading to a more sustainable waste management system (M. Sharma et al., 2022). Furthermore, FW can be converted into animal feed but converting FW into animal feed faces challenges such as ensuring safety from contaminants, meeting nutritional requirements, adhering to regulatory standards, optimizing processing methods, and addressing market acceptance concerns. These factors require careful management and technological innovation to safely and effectively utilize food waste in animal feed applications (Nath et al., 2023; Rajeh, Saoud, Kharroubi, Naalbandian, & Abiad, 2021). Furthermore, a novel approach involves using insects to break down food waste and produce valuable products like protein-rich insect-based foods. For example, companies are exploring the use of black soldier fly larvae to convert FW into protein powder and animal feed. However, despite its sustainability benefits, consumer acceptance remains a challenge due to cultural barriers and perceptions about insect consumption in many regions and religions (Bhatia, Jha, Sarkar, & Sarangi, 2023; Roy, Mohanty, Dick, & Misra, 2023).

6.1. Physical pre-treatment techniques

Food waste encompasses mechanical and thermal methods, such as milling, ultra-sonication, Pulsed-electric-field (PEF) technology, microwave irradiation, and extrusion. Mechanical methods, like milling, enhance anaerobic digestion by increasing the substrate's surface area.

Milling exposes food waste to significant mechanical stress, leading to changes in its physicochemical and structural characteristics (Gallego-García, Moreno, Manzanares, Negro, & Duque, 2023). This process improves enzymatic action and digestibility by enlarging the surface area of the pretreated material. This method is frequently combined with other pre-treatment methods, such as wet milling, to extract cellulose nano-fibers or sugars from lingo-cellulosic materials (Gallego-García et al., 2023).Despite its benefits, milling has limitations, including high energy requirements and, in some cases, comparatively lower efficiency. Additionally, PEF technology increasingly researched in the past decade, applies voltage to food waste to extract bio-compounds, particularly from fruit and vegetable waste (Arshad et al., 2021). This method triggers irreversible electroporation in biological cells, disrupting cell membranes and enabling the extraction of valuable compounds. PEF technology is advantageous for sustainability, as it requires less energy and is effective in various industries. For instance, improper handling of food processing wastes emits CO2, contributing to the greenhouse effect and necessitating energy and cost-intensive waste disposal methods. Therefore, eco-friendly and cost-effective approaches such as PEF technology are recommended for extracting bioactive components from these wastes. The extracted compounds exhibit higher antioxidant activity, stability, therapeutic, and functional properties, making them suitable for use in the food and pharmaceutical industries (Theagarajan, Balendran, & Sethupathy, 2024; Yan et al., 2010). PEF, combined with conventional or non-conventional technologies like ultrasound (US) and high-pressure processing (HPP), effectively extracts bioactive compounds with noticeable advantages such as limited solvent consumption, high purity, reduced extraction time, and low energy consumption (Ashrafudoulla et al., 2023). Compared to Ultrasound Assisted Extraction (UAE), Microwave Assisted Extraction (MWAE), and HPP-assisted extraction, PEF-assisted extraction (PEFAE) consumes less energy and provides a higher extraction yield, making it a promising technology for valorizing food processing wastes and supporting a circular economy. However, challenges include treatment optimization, installation costs, and electrochemical changes in electrodes. Future studies need to focus on valorizing wastes from dairy, spices, and condiment industries. Although PEF technology has broad applications in waste valorization, the initial cost of the PEF system and the selection of appropriate solvents are major concerns. Additionally, optimizing PEF treatment conditions is necessary to achieve high extraction yield and product quality in a short time (Velusamy, Rajan, & Radhakrishnan, 2023).

Furthermore, Extrusion as a pre-treatment technique, reduces particle size and fibrillates materials through a combination of mechanical and thermal effects in an extruder (Gallego-García et al., 2023). A continuous operation and better mixing are among its advantages, making it suitable for extracting value-added bio-products or promoting sugar yield from various food waste sources. These physical pre-treatment methods enhance microbial and enzymatic hydrolysis of FW by increasing surface area and disrupting complex compounds (Konan et al., 2022).

Table 3
Valorization of food wastes into biofuels using microbial processes.

Type of FW	Pre-treatment	Microorganisms used	Biofuel produced	References
Kitchen FW and Rice Straw Hydrolysate	*Clostridium butyricum* CBT-1		358.15, 300.8, and 294.5 NmL/gSub from glucose, starch, and cellulose, respectively	Shah et al. (2023)
Bread waste	2% HCl (v/v)	S.*cerevisiae* strain KL17	114.9 g/L	Narisetty et al. (2022)
Potato Peel Waste (PPW)	Ethanol-50-75% (v/v); Temp:120–180 °C; 1% (w/w) H2SO4 1% NaOH	*Aspergillus fumigatus* (for production of enzymes) and S. *tanninophilus*	9.451 g/L	Jin, Song, and Liu (2020)
Fruits and vegetable processing waste	Autoclaving 120 °C; 15 psi; 20min	Dark Dry Fermentation Temp:55 ± 1 °C; SS:250 rpm; pH:5.5 to 6.75	41%	Abubackar et al. (2019)
FW	Lactic acid fermentation		94.6 ± 25.11 mL/gVS	Roslan et al. (2023)

A study by Yue et al., explored the use of ultrasound and microwave pre-treatment techniques to enhance methane production from lipid-rich FW. Ultrasound pre-treatment at 50,000 kJ/kg-TVTS resulted in a higher soluble COD of 10,130 mg/L compared to 1910 mg/L with microwave heating at the same energy input. Methane yield from ultrasound-pretreated waste increased by 43.3%–927.97 mL/g-TVTS, surpassing the 738.63 mL/g-TVTS yield from microwave heating. Fourier transform infrared spectrometer (FTIR) and scanning electron microscopy (SEM) analyses showed reduced residual lipids and relieved microorganism coating after ultrasound pre-treatment. The energy conversion efficiency to methane was 69.89% for ultrasound-treated waste, higher than the 58.98% achieved with microwave heating. These findings underscore ultrasound's effectiveness in degrading lipids and enhancing energy recovery through methane production, positioning it as a promising strategy for FW upcycling (Yue et al., 2021). Moreover, hydrothermal processing, involving hot compressed water, can convert food waste into valuable chemicals or materials, like converting market fruit and vegetable waste into hydro-char for soil improvement (Zhang, Qin, Sun, & Wang, 2022).

6.2. Chemical pre-treatment methods

Chemical pre-treatment methods involve treating food waste with acids, alkalis, oxidative agents, or ionic liquids. These agents can break down complex organic materials, making subsequent processing steps more efficient. For instance, acids and alkalis can hydrolyze lingo-cellulosic materials, while oxidative agents can degrade stubborn organic compounds. Furthermore, ionic liquids are gaining attention for their ability to selectively dissolve and recover valuable components from food waste, contributing to a circular economy (Eqbalpour et al., 2023). From a practical perspective Pakistan, a developing country grappling with energy shortages, has abundant sources of wasted parthenium biomass and catering waste (McElroy, 2018). Thermochemical conversion methods offer an efficient means to convert food waste into valuable fuels and carbon materials, yielding enegy. For example, in the forthcoming decades, hydrogen gas produced by thermochemical process is anticipated to ascend as the preeminent sustainable fuel, owing to its substantial energy density of approximately 122 MJ/kg and its emission-free combustion, thereby positioning it as a cornerstone of the carbon-neutral economy. This phenomenon has precipitated a notable surge in global hydrogen production and storage, with an estimated annual sales potential of €630 billion by 2050, capable of meeting up to 24% of the world's energy demand (El-Qelish, El-Shafai, Azouz, Rashad, & Elgarahy, 2024). For instance, acid pre-treatment entails treating lignocellulose-rich food waste with potent acids like nitric acid, hydrochloric acid (HCl), or sulfuric acid (H2SO4), along with other acids like acetic acid, maleic acid, or citric acid. These strong acids can dissolve cellulose, lignin, and hemicellulose, but they might also generate inhibitory compounds, necessitating pH optimization for effective pre-treatment (Elgarahy et al., 2023a; Sołowski, Konkol, & Cenian, 2020). Diluted sulfuric acid (H2SO4) has been utilized to hydrolyze different materials, such as soybean straw and mixtures of food waste. Moreover, subjecting substances to pre-treatment with high-pressure carbon dioxide can trigger the hydrolysis of hemicellulose. Additionally, Ionic liquid pre-treatment involves the use of organic salts with high thermal stability to dissolve cellulose, thereby improving the efficiency of hydrolysis. Various types of ionic liquids have been applied to different FW substrates, enhancing their enzymatic digestibility and fermentation efficiency. However, this method is costly despite its effectiveness (Elgarahy et al., 2023a; Sołowski et al., 2020).

Furthermore, alkaline pre-treatment is a process that involves treating FW rich in cellulose with alkaline mediums such as sodium hydroxide (NaOH). This treatment aims to separate lignin and hemicellulose from cellulose, thereby modifying the structure of cellulose to enhance hydrolysis and improve thermodynamic stability (Jankovičová, Hutňan, Czölderová, Hencelová, & Imreová, 2022). However, it is crucial to optimize the concentration of alkali to avoid toxicity or inhibition of microbial growth. Although alkaline pre-treatment has been applied to various substrates, it can prolong hydrolysis duration and lead to the formation of salts. Oxidative pre-treatment is a process that entails treating lingo-cellulosic material with alcohol-acidic solutions, peroxides, or potent oxidizing agents such as potassium permanganate. This treatment aims to separate hemicellulose from cellulose and dissolve lignin (Fisgativa, Saoudi, & Tremier, 2016). While effective, oxidative pre-treatment is costly and may not substantially improve saccharification efficiency.

6.3. Biological pre-treatment methods

Biological approaches present a sustainable remedy, as microorganisms can be harnessed to generate biofuels, electricity, bio-surfactants, bio-plastics, bio-fertilizers, and other valuable products (P. Sharma, Gaur, Kim, & Pandey, 2020). For instance valorization of food wastes into biofuels using microbial processes is presented in Table 3. Enzymatic treatment breaks down proteins into amino acids and carbohydrates into sugars. Fungal treatment releases enzymes such as cellulase and hemicellulases, which further break down organic matter. This synergistic effect often produces additional valuable by-products (Elgarahy et al., 2023a; Maurya, Kumar, Chaurasiya, Hussain, & Singh, 2021). Furthermore, Black Soldier Fly Larvae (BSFL) can be used to convert restaurant food scraps into high-protein feed for animals. To upcycle citrus peels, the extraction and utilization of pectin can be effectively achieved through enzymatic hydrolysis. This methodology allows for the efficient breakdown of citrus peel cell walls, releasing pectin for subsequent use. The extracted pectin can then be incorporated into various food products such as baking, confectionery, and meat products as a natural gelling agent. Additionally, it can be marketed as a functional food ingredient to enhance gut health by supporting beneficial gut microbiota. It not only maximizes the value of citrus peels but also aligns with sustainable practices by repurposing food waste into valuable ingredients.

Moreover, anaerobic digestion is a widely accepted method for

decomposing biodegradable organic food waste in controlled atmospheric conditions, resulting in biogas rich in methane, carbon dioxide, and trace gases (Uddin et al., 2021). Various purification methods, such as chemical absorption, and membrane separation, help remove impurities, enhancing the energy value of biofuels. Furthermore, coffee waste can be effectively upcycled into biofuels such as biogas and bioethanol through anaerobic digestion and fermentation processes. Anaerobic digestion breaks down organic matter in coffee waste by microorganisms in the absence of oxygen, producing biogas, which can be used for heating, electricity, or as vehicle fuel. For example, 100 tons of coffee waste can generate approximately 4000 cubic meters of biogas, providing a renewable energy source. Fermentation converts carbohydrates in coffee waste into bioethanol using yeasts like *Saccharomyces cerevisiae*, producing ethanol as a biofuel. This method can yield around 30 L of bioethanol from one ton of coffee waste, contributing significantly to sustainable waste management and energy production. Additionally, FW rich in carbohydrates can serve as a viable substrate for producing ethanol, hydrogen, and succinic acid, offering potential solutions for greenhouse gas reduction and sustainable energy generation (Papadaki & Mantzouridou, 2019; Song et al., 2021).

6.4. Fermentation and hydrolysis

Organic acids, including citric acid, succinic acid, lactic acid, and acetic acid, can be obtained from FW and have diverse applications in industries such as food, pharmaceuticals, and cosmetics. Through innovative conversion methods like fermentation and enzymatic hydrolysis, organic acids can be produced from waste materials, contributing to a more sustainable environment (Behera, Mishra, & Mohapatra, 2021; Harun, Hassan, Zainol, Ibrahim, & Hashim, 2019). For example, a study by Papadaki et al., aimed to optimize an integrated simple process for citric acid production using Spanish-style green olive processing wastewaters enriched with sugars from white grape pomace and the robust Aspergillus niger B60. Mild mixing of equal quantities of the above streams governed satisfactory amount of appropriate carbon sources (equimolar mixture of glucose and fructose, 111.5 g/L) in the sugar-enriched wastewater and its neutralization. Various nutrients and fermentation conditions were investigated and maximum citric acid content (85 g/L) and yield (0.56 g/g) were obtained in liquid surface culture after minimum regulation by adding sucrose and NH_4NO_3 (100 g/L and 1.1 g/L, respectively). Scale-up experiments (5 L-scale) verified findings from small scale (250 mL). The chemical oxygen demand value and phenolic content of the treated wastewater were reduced by 78% and 64%, respectively. Findings support the potential for clustering the respective enterprises in a biorefinery plant for citric acid fermentation. Likewise, lactic acid can be effectively generated from restaurant and bakery waste using enzymatic hydrolysis and fermentation processes, providing environmentally friendly alternatives to chemical methods. Additionally, advanced fermentation techniques use specific microbial strains to ferment food waste into high-value products, such as fermenting spent grain from breweries to create probiotic beverages or flavor-enhancing ingredients (Papadaki & Mantzouridou, 2019).

A study by Chouaibi et al., revealed that pumpkin peel waste, rich in starch, can be effectively converted into bioethanol. This process involves optimizing reducing sugar concentration and bioethanol production using artificial neural networks (ANN) and response surface methodology (RSM). The hydrolysis process conditions are: 120 min hydrolysis time, 17.5 g/L substrate loading, 7.5 U/g α-amylase, and 56.40 U/mL amyloglucosidase. For fermentation, the optimal conditions are: 45 °C temperature, pH 5.06, 188.5 rpm shaking speed, and 1.95 g/L yeast concentration. Under these conditions, the experimental reducing sugar and bioethanol concentrations are 50.60 g/L and 84.36 g/L, respectively. The ANN model proved superior in prediction accuracy, and key factors influencing production were substrate loading and fermentation temperature. This method highlights pumpkin peel waste as a viable source for fuel-ethanol production (Chouaibi, Daoued, Riguane, Rouissi, & Ferrari, 2020). Furthermore, frozen FW, especially those with high potassium levels, can be upcycled into commercial alcohol production through the fermentation process. This method utilizes the mono-, di-, polysaccharides, and starch present in the waste. During fermentation, microorganisms such as yeast convert these sugars into ethanol, a valuable commercial alcohol. The ideal conditions for ethanol fermentation occur within a pH of 4.5–5.5 and a temperature range of 20–35 °C. For example, one ton of such waste can yield approximately 100 L of ethanol, providing an efficient way to repurpose frozen FW into a useful product while reducing environmental impact (Bangar, Chaudhary, Kajla, Balakrishnan, & Phimolsiripol, 2024; Banu, Kumar, Gunasekaran, & Kavitha, 2020).

7. Food waste upcycling into valuable products

7.1. Packaging materials

A study by Gupta et al., revealed that addressing the dual challenges of plastic waste and FW in populous regions like India and China requires innovative solutions. In India alone, major cities generated approximately 26,000 tonnes of plastic waste daily in 2019–20, with only 60% being recycled. Concurrently, these countries produced vast amounts of food waste, with China and India accounting for 91 and 69 million tonnes respectively in 2019. To mitigate these issues, biodegradable packaging films offer a promising avenue. By utilizing biocompounds such as proteins, cellulose, starch, and lipids derived from food and vegetative waste, researchers are developing films that can replace non-biodegradable plastics. Despite challenges in mechanical strength and barrier properties, advancements in material science aim to optimize these films for practical packaging applications, thereby reducing environmental pollution and enhancing sustainability efforts (Gupta, Toksha, & Rahaman, 2022). Furthermore, edible films made from milk whey proteins, pectin from citrus peels, and chitosan from shrimp shells have been shown to extend the shelf life of fresh produce by protecting it from environmental conditions and reducing spoilage. These advancements not only contribute to waste reduction but also enhance the overall sustainability of food service operations by utilizing food waste to create valuable packaging materials (Nunes et al., 2023).

7.2. Bio-pigments pyocyanin and 1-hydroxyphenazine

Phenazines, such as pyocyanin and 1-hydroxyphenazine, are pigments found in *Pseudomonas aeruginosa*, contributing to its blue-green color. These compounds play various roles, including being electrochemically active and beneficial for the host. Additionally, they possess antimicrobial and anticancer properties. However, their use is restricted due to the high cost of carbon substrates. Utilizing carbon derived from FW in bio refineries could help overcome this limitation (Pantelic et al., 2023). Moreover, waste materials like vegetable peels, fruit scraps, coffee grounds, leftover food, and used cooking oil can be collected and processed through hydrolysis, fermentation, and pyrolysis or gasification to extract carbon. This not only provides a cost-effective carbon source but also promotes sustainable waste management, facilitating the production of valuable phenazines in biorefineries (Ganesh, Sridhar, & Vishali, 2022; Verma, Rao, Joshi, Choudhary, & Srivastava, 2022).

7.3. Super activated hydro-char for remarkable hydrogen storage

According to a study the simulated FW, including apple, bread, green beans, cabbage, cheese, and canned chicken, underwent hydrothermal carbonization at 220 °C and chemical activation at 800 °C using various KOH-to-hydrochar ratios (2:1, 3:1, 4:1). The resulting solid products, termed super-activated hydro-chars, were examined for surface properties using XRD, nitrogen adsorption isotherms at 77K, and proximate and ultimate analyses. The super-activated hydro-chars exhibited surface areas ranging from 2070 to 2885 m^2/g and total pore volumes of

0.98–1.93 cm^3/g. Notably, one sample produced with a 4:1 KOH-to-hydrochar ratio demonstrated a hydrogen storage capacity of up to 6.15 wt% at 23 bar and 77 K, showcasing the potential of up-cycled FW for effective hydrogen storage. It can be concluded that FW is a suitable source for developing hydrogen gas adsorbents through hydrothermal carbonization and KOH activation. Higher hydro-char-to-KOH ratios improved material porosity, with optimal properties achieved at 4:1. The hydrogen adsorption capacity surpassed the US Department of Energy's 2020 target, highlighting food waste's promise for hydrogen storage material development (Saha & Reza, 2021). While the study demonstrates the potential of food waste-derived super-activated hydro-chars for hydrogen storage, a notable gap exists in the long-term stability and reusability of these materials. Future research needs to focus on assessing the durability and performance of these hydro-chars over multiple adsorption-desorption cycles to ensure their practical applicability in real-world hydrogen storage systems.

7.4. Polyunsaturated fatty acids (PUFA) production

An adaptive evolution process involving two factors significantly increased biomass and PUFA production in wild-type *P. tricornutum* under high-glucose conditions, resulting in the creation of an evolved strain, ALE-Pt1. This strain efficiently utilized food waste hydrolysate as an affordable carbon source during cultivation under mixotrophic conditions, leading to improved biomass yield and PUFA production (Wang et al., 2022). Furthermore, a study conducted using upcycling food waste valorization methods revealed the potential of single-cell protein technology to address global protein shortages, highlighting its independence from climate, soil characteristics, and available land (Hülsen, Hsieh, Lu, Tait, & Batstone, 2018).

7.5. Transforming food waste into medical product components

A study by Carolo et al., created a marine collagen scaffold by combining 2 g/L of collagen with 0.01% TritonX-100, which was then placed into rubber silicon molds, frozen overnight, and subsequently lyophilized overnight. To test the scaffold, researchers inflicted a wound on the dorsal column of adult male rats and applied the scaffold to the wound area. When compared to a commercially available dermal regeneration product, the marine collagen scaffold demonstrated comparable biocompatibility, promoted angiogenesis, and facilitated the deposition of mature collagen (Carolo et al., 2023). Rajbimashhadi et al., emphasized the methods for extracting collagen from fish industrial wastes and its applications in tissue engineering and wound healing (Moffat, Ye, & Jin, 2022). Furthermore, food waste-derived materials have been investigated for fabricating nanoparticles for drug delivery. For example, Dai et al., extracted alkali lignin from corn cob using hydrothermal treatment to create nanoparticles with an organic solvent solution. These nanoparticles were then combined with resveratrol and magnetite to form a nano-drug carrier. The resulting materials demonstrated good stability, biocompatibility, and a relatively high drug loading capacity of over 20% by weight. Additionally, they effectively inhibited tumor growth and improved survival rates in experimental animals (Dai, Liu, Hu, Zou, & Si, 2017). In another study, researchers integrated citrus pectin into a copper-based metal–organic framework with folic acid. The resulting fibers exhibited antibacterial activity, biocompatibility, and good tensile strength (Kiadeh et al., 2021). Research in these areas is crucial as it offers innovative solutions for waste valorization, turning food industry by-products into valuable medical and pharmaceutical materials. These advancements not only promote sustainability by reducing waste but also provide new avenues for developing biocompatible and efficient therapeutic options. Continued exploration and optimization in this field can significantly enhance the effectiveness and applicability of biomedical products, contributing to improved health outcomes and environmental conservation.

8. Case studies

8.1. Food waste reduction in Swedish public catering

An analysis of eight years of FW data from Swedish public catering was conducted to assess progress towards the global goal of halving FW by 2030. The findings indicated a 15–30% reduction, suggesting a downward trend. If all canteens perform similarly, achieving the 2030 target appears feasible. However, it's possible that these canteens represent the best performers, which could lead to an underestimation of the actual change or current waste levels. Swedish preschools and schools generate 19,000–21,000 tons of FW annually, highlighting the need for waste monitoring tools, progress tracking mechanisms, and incentives for further reduction (Malefors, Strid, & Eriksson, 2022b). The study observed decreasing FW levels and trends across all sectors of Swedish public catering over the study period. By 2020, primary schools reduced waste by 16%–42 g per portion, preschools by 26%–53 g, secondary schools by 20%–66 g, and elderly care homes by 43%–56 g. Primary school data had the highest representativeness, impacting the overall results. Scaling up, FW reduction in the sector from 2016 to 2020 ranged from 15% to 30%, depending on the portion factor used. In 2020, Swedish school catering generated 19,000 to 23,000 tons of FW. Achieving a 50% reduction is feasible with current trends and by helping underperforming canteens improve. Understanding the causes of food waste is a key, and canteens need tools for waste reduction and progress tracking. Verification of data through random samples could improve accuracy, and legal procedures may ensure data provision from establishments not addressing the issue (Malefors, Strid, & Eriksson, 2022b).

8.2. Food waste from Air-line catering

The airline industry's food provisioning practices often lead to unsustainable consumption and FW (You, Bhamra, & Lilley, 2020). In 2018, airlines worldwide disposed of approximately 6.1 million tons of cabin waste, with approximately 20–30% of this total, around 2.1 million tons, comprising unused food and beverages. According to the International Air Transport Association, managing this 5.7 million tons of cabin waste incurred costs totaling approximately US$927 million for the aviation sector (Elwakeel, Elgarahy, Alghamdi, & El-Qelish, 2023). A study focusing on Halal food production identified and analyzed FW hotspots in a flight catering. The research aimed to assist organizations in developing FW management policies for sustainable growth. Vegetable waste accounted for 40–50% of the total FW generated during operations. Efforts to reduce FW in airline catering should prioritize the reduction of seafood waste, as this can lead to significant cost savings. This can be achieved through menu and recipe design, as well as engaging with customers to understand their preferences and consumption patterns better. By aligning food offerings with customer preferences and optimizing portion sizes, airlines can reduce overproduction and minimize waste (Halizahari, Mohamad, Anis, & Wan, 2021). Future research in this area could explore the key factors driving waste generation in airline catering businesses, including clients, suppliers, and staff. Understanding these factors can help develop more targeted waste reduction strategies. For example, identifying specific menu items that are consistently wasted can help tailor menu designs to minimize waste. Similarly, understanding how suppliers' practices impact waste generation can help airlines work with their suppliers to implement more sustainable practices. Engaging with staff to raise awareness about the importance of reducing FW and providing them with the tools and training to minimize waste can also be effective strategies. By involving all stakeholders in the food provisioning process, airlines can create a culture of sustainability that extends throughout their operations. The airline industry faces significant challenges in managing FW, but there are opportunities for improvement. One example of successful FW reduction in the airline industry is the initiative undertaken by "Delta Air Lines". Delta implemented a

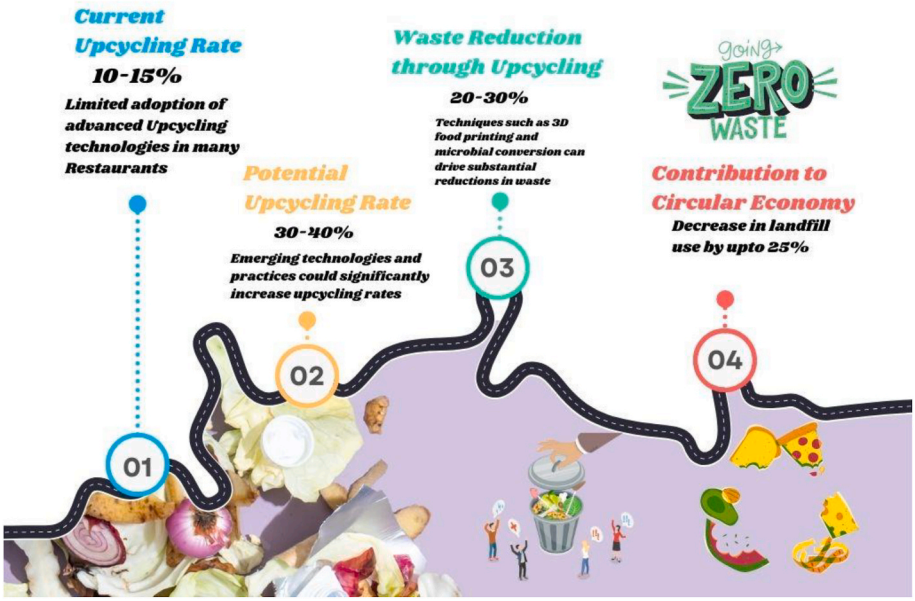


Fig. 4. The power of upcycling: Current success and future potential.

comprehensive FW management program that included measures such as menu planning based on customer preferences, optimizing portion sizes, and donating surplus food to local charities. Additionally, Delta partnered with composting and recycling facilities to ensure that unavoidable food waste was repurposed sustainably (report, 2024). As a result of these efforts, Delta significantly reduced its FW, demonstrating that strategic waste reduction and upcycling methodologies can lead to both environmental benefits and cost savings. Therefore, by focusing on preventing and reducing waste at the source, airlines can minimize their environmental impact and achieve cost savings. Future research in this area needs further exploration of innovative strategies for reducing food waste in airline catering, taking into account the unique challenges and opportunities of the industry.

8.3. Quick-service restaurants (QSRs) on a university campus

A study validated a tool for quantifying pre-consumer food waste in QSRs and used statistical process control to map the waste generation process. The study also investigated FW categories and factors, such as menu practices, that influence waste production. Food waste audits were carried out in two Australian university campus food outlets, each for duration of two weeks, using direct weighing. Observations and menu practices were documented and compared with reports from owner-managers. Waste quantities were analyzed using statistical process control. Both outlets had an average daily pre-consumer food waste of around 25 kg, with 60% being inedible waste, totaling about 5.5–6.1 kg per full-time-equivalent employee (FTE) per day. Coffee grounds were the largest contributor to total waste. Both outlets were compliant with Australian best practices (Jayasekara, McGrath, Kravchuk, Zhou, & Morris, 2024). The study concluded that the method for separating and quantifying pre-consumer food waste was feasible for the QSRs under investigation. By employing statistical process control, the study was able to distinguish between common and special causes of variation, providing valuable insights for food waste reduction initiatives (Jayasekara et al., 2024). But the study's brief audit period (9–10 days) restricted the identification of patterns. Extended audits could uncover trends, assisting in FW management. Further research involving QSRs is necessary to comprehend FW generation and enhance processes. Identifying coffee grounds as a significant waste stream could inform targeted reduction strategies. The tool and methodology are valuable for evaluating the stability of FW production processes, identifying hotspots, and detecting anomalies that could affect FW generation. This information can help guide FW reduction strategies and the development of best practices (Jayasekara et al., 2024).

8.4. Successful initiatives of food waste reduction

The current success rate and potential of upcycling FW is given in Fig. 4. Furthermore, in a study by Beretta et al., 13 cases of initiatives aimed at reducing FW were analyzed. Avoidable FW amounts to 108 g per meal, which was 13% of the food purchased, resulting in 238 g of CO2-equivalent emissions per meal. Reduction rates observed in the case studies ranged from 32% in the education sector to 62% in the business sector. On average, a 38% decrease in FW correlated with a 41% decrease in climate impacts and a 30% decrease in biodiversity impacts. In a more extensive reduction plan, food services made use of 50% of non-marketable vegetables that would otherwise be discarded, leading to a 70% reduction in food waste (Beretta & Hellweg, 2019).

9. Challenges

Implementing upcycling practices in restaurants presents several notable challenges. Staff training is a major hurdle, with 60% of restaurant managers reporting difficulties in effectively training staff to handle upcycled ingredients, leading to potential food safety issues. Supply chain adjustments also pose significant problems; 50% of restaurants struggle with sourcing consistent upcycled materials, indicating issues with logistics and cost management. Maintaining food quality further complicates efforts, as 30% of restaurants must alter their menus or cooking processes, with 25% of customers expressing dissatisfaction with upcycled dishes. Consumer acceptance remains a complex challenge due to cultural barriers and perceptions about insect consumption, which hinder methods like using black soldier fly larvae. In British and Dutch restaurants, while effective strategies like demand forecasting and passive disposal are utilized, proactive measures such as ingredient repurposing and reducing plate waste are less common due to organizational and regulatory barriers. Additionally, consumer perceptions of food packaging impact waste reduction efforts, highlighting a research gap on packaging's role in waste mitigation. Emerging technologies introduce further issues, including high installation costs and electrochemical challenges with PEF, uncertainty about the stability and reusability of food waste-derived super-activated hydro-chars, and

Table 4
Summarized challenges for Food waste management and their potential solutions.

Challenges	Potential solutions	References
Lack of Consumer Awareness	Implement educational campaigns and labels to raise awareness.	Di Talia, Simeone, and Scarpato (2019)
Logistical Challenges	Use technology to optimize collection routes and schedules.	Jagtap et al. (2020)
Regulatory Compliance	Implement robust tracking and documentation systems.	George, Harsh, Ray, and Babu (2019)
Storage Limitations	Invest in compacting or dehydrating systems to reduce volume.	Shiekh (2020)
Cost Constraints	Collaborate with food banks or local governments for cost-sharing.	Iacovidou, Hahladakis, and Purnell (2021)
Supply Chain Fragmentation	Establish partnerships and networks to streamline processes.	Iacovidou et al. (2021)
Technological Barriers	Develop open-source or affordable technologies.	Bangar et al. (2024)
Staff Training	Provide regular training and incentives for staff.	Al-Kandari, Al-abdeen, and Sidhu (2019)
Perception of Quality	Implement quality control measures and communicate benefits to customers.	Al-Kandari et al. (2019)
Waste Sorting	Provide clear guidelines and tools for sorting.	Jamali and Misman (2021)

variable performance of machine learning models for catering demand prediction. The scalability of microbial processes and ensuring consistent quality in 3D food printing of upcycled waste also presents critical challenges. However, Table 4 presents these issues and their potential solutions. Addressing these issues requires targeted solutions, such as comprehensive training programs, robust supply chain partnerships, quality control measures, clearer regulatory guidelines, and consumer education, with tailored approaches needed to effectively overcome barriers and advance upcycling practices in the restaurant industry.

10. Conclusions and future perspectives

In conclusion, this work highlights the urgent need for effective upcycling of food waste in the restaurant and catering sectors. Although progress has been made in repurposing leftovers and implementing waste reduction strategies, significant challenges persist. The existing literature reveals that while various methods for food waste upcycling, such as enzyme treatments and microbial processes, have been explored, a comprehensive and integrated approach remains elusive. Future research needs to focus on critically assessing the current upcycling methods and their practical applications. Comparative studies are needed to evaluate the cost-effectiveness, scalability, and environmental impact of these technologies. For instance, advancements in 3D printing technology present opportunities for creating high-value edible products from food waste, which warrants further exploration. Another crucial area for research is understanding consumer perceptions of food packaging and its role in waste reduction. Investigating how packaging influences consumer behavior can lead to the development of solutions that reduce waste and enhance the appeal of upcycled products. The application of machine learning for menu prediction and meal customization also offers promising avenues for reducing waste in catering operations. Future studies need to refine these models to improve their accuracy in forecasting demand and minimizing FW. Additionally, long-term research is needed to assess the stability and performance of food waste-derived products, such as super-activated hydro-chars used in hydrogen storage. Ensuring these products' practical applicability requires rigorous testing over extended periods. Scaling up successful microbial processes, like those involving strains such as ALE-Pt1, necessitates optimizing performance and integrating these processes into existing waste management systems. Overcoming technical and economic barriers to widespread adoption is crucial for this advancement.

Case studies, such as Delta Air Lines' comprehensive food waste management program, demonstrate the potential for significant waste reduction. Future research needs to apply these findings to other sectors, like quick-service restaurants, to develop targeted waste reduction strategies. Identifying and addressing key waste streams in various sectors can lead to more effective and tailored waste management practices. Finally, collaboration with industry stakeholders and the integration of advanced technologies will be key to enhancing the effectiveness of waste management strategies. Fostering partnerships between researchers, industry practitioners, and policymakers can drive innovation and facilitate the implementation of sustainable solutions. Ultimately, advancing food waste upcycling requires a combination of critical evaluations of current methods, exploration of new technologies, and an understanding of consumer behavior. By addressing these areas, the field can move towards more sustainable and practical solutions for managing food waste.

Data availability

Data will be made available on request.

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